



BE Candidate Take Home Test

Have you shopped online? Let's imagine that you need to build a checkout backend service that will support different promotions with the given inventory.

Build a checkout system with these items:

 SKU	 Name	 Price	 Inventory Qty
120P90	Google Home	\$49.99	10
43N23P	MacBook Pro	\$5,399.99	5
A304SD	Alexa Speaker	\$109.50	10
234234	Raspberry Pi B	\$30.00	2
 New			

The system should have the following promotions:

- Each sale of a MacBook Pro comes with a free Raspberry Pi B
- Buy 3 Google Homes for the price of 2
- Buying more than 3 Alexa Speakers will get a 10% discount on all Alexa speakers

Example Scenarios:

Scanned Items: MacBook Pro, Raspberry Pi B

Total: \$5,399.99

Scanned Items: Google Home, Google Home, Google Home

Total: \$99.98

Scanned Items: Alexa Speaker, Alexa Speaker, Alexa Speaker

Total: \$295.6



Evaluation Checklist

This exercise's functional correctness is secondary because it is so simple. Determining accurate outputs should be an obvious accomplishment. In order to ensure that your work truly stands out, we consider the following:

Code quality & readability: Will any random engineer be able to understand the execution just by briefly scanning through the source code? Software design: Does the implementation make full use of classes, objects, functions, abstractions, interfaces, etc.

Engineering best practices: Does it follow proper architectural patterns (like MVC) and SOLID principles?

Any automated tests (e2e, integration, unit, etc.)

Submission

We recommend spending +6 hours in total. Once you have completed the exercise, please push the git repository to a host of your choice, preferably GitHub. Your Dockerfile and code should be sufficient for us to recreate and test your API.

Please submit the following items:

Git repository for your code (including Dockerfile), database design documents (DB structure and explanation) and CI (a plus point)